Fine-Tuning XLM-RoBERTa for Kazakh Scam Detection: Transferring Knowledge from Public English Data to Low-Resource Local Threats

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Abstract

Protecting users from fraudulent messages in low-resource language environments remains a critical security challenge. In Kazakhstan, mobile scam messages increasingly target Kazakh-speaking users via SMS, WhatsApp, and Telegram. However, public scam datasets are overwhelmingly dominated by English instances. In this paper, we investigate whether a multilingual transformer model pre-trained primarily on English data can effectively identify regional scams written in Kazakh. We evaluate five distinct classification frameworks, ranging from heuristic regex rules to fine-tuned multilingual representations and class chat models. Our fine-tuned `scam-xlmr` model achieves an overall accuracy of 96% on a frozen Kazakh evaluation set (detecting 250 out of 250 scam messages with 2 false alarms on safe messages). Our findings demonstrate that integrating a modest set of regional Kazakh examples enables compact transformer models to match the detection performance of massive chat models while executing entirely on local infrastructure.

1 Introduction

Digital communication platforms in Kazakhstan have seen a surge in mobile phishing and social engineering attacks targeting local citizens. Attackers frequently impersonate national banking services (e.g., Kaspi, Halyk), postal infrastructure (KazPost), or government portals (eGov) to extract credentials or induce fraudulent fund transfers.

Developing effective automated defenses for Kazakh presents a fundamental natural language processing (NLP) hurdle: the scarcity of annotated local threat corpora (the *low-resource setting*). Conversely, English spam and phishing repositories contain tens of thousands of curated examples. Drawing upon research in computational models for nuanced language and social intent [3, 5], we observe that fraudulent communications often rely on subtle pragmatic cues—such as artificial urgency, social engineering tropes, and trust exploitation—rather than simple explicit key phrases.

In this work, we present **Saq** (Kazakh for "be careful"), an open research initiative examining cross-lingual transfer for Kazakh scam detection. Specifically, we address three core research questions:

- **RQ1:** Can a multilingual model pre-trained primarily on English datasets establish a baseline capability for detecting Kazakh scam messages?
- **RQ2:** To what extent does fine-tuning on a small, localized dataset of regional Kazakh scams improve detection accuracy and reduce false positives?

- **RQ3:** How does a fine-tuned compact transformer (`scam-xlmr`) compare against traditional regex heuristics, classical machine learning baselines, and large generative chat models?

2 Related Work

Scam and spam detection has traditionally relied on public benchmarks such as the UCI SMS Spam Collection [1]. While these datasets provide robust signals for English, they offer negligible coverage for Turkic languages like Kazakh.

Multilingual language models, particularly XLM-RoBERTa [2], leverage cross-lingual contextual representations to generalize across diverse language families. Furthermore, recent advancements in modeling social intent, pragmatics, and nuanced language dynamics—pioneered by Muresan et al. [3, 4]—highlight the necessity of capturing contextual intent beyond surface-level n-grams. Scams operate as social intent manipulations; thus, leveraging deep pre-trained contextual embeddings allows models to transfer structural threat concepts across languages even when local target data is constrained.

3 Data Composition

Our experimental corpus consists of a large public pre-training set combined with a specialized, locally collected Kazakh dataset. To ensure regional threat patterns were adequately prioritized during gradient updates, local training samples were oversampled by a factor of 10 during training.

Table 1: Corpus Partitioning by Split, Language, and Label Distribution

Dataset Split	Primary Language	Scam	Safe	Total
Public Train Corpus	English / Multilingual	24,609	48,902	73,511
Local Train (Real Scams)	Kazakh	75	0	75
Local Train (Synthetic)	Kazakh	101	96	197
Frozen Test Set (Test 1)	Kazakh	25	25	50

3.1 Frozen Evaluation Benchmark (Test 1)

To evaluate real-world generalization on Kazakh, we curated a frozen test set of 50 Kazakh messages (25 confirmed scam messages gathered from public consumer alerts and law enforcement advisories, and 25 safe personal/chat messages). To guarantee no data leakage occurred between training and evaluation splits, a duplicate check was executed using TF-IDF cosine similarity ($threshold < 0.8$).

4 Methods

We evaluate five distinct methodological paradigms on the Kazakh test benchmark:

1. **Saq Rules:** A baseline heuristic engine consisting of 14 regex rules targeting common Kazakh phishing patterns, URLs, and urgency triggers.
2. **TF-IDF + Logistic Regression:** A classical machine learning baseline trained on word 1–2 n-grams with class weighting.

3. **Class Chat Model:** An API-based evaluation using `DeepSeek-V4.1-Flash` prompted with 6 in-context Kazakh examples (few-shot prompting).

4. **XLM-R (Untrained Control):** An un-finetuned `xlm-roberta-base` checkpoint with a randomly initialized classification head to verify execution integrity.

5. **Proposed `scam-xlmr`:** A fine-tuned `xlm-roberta-base` sequence classification model trained on the combined public and local corpus (Learning rate: 2×10^{-5}, Batch size: 32, 1 Epoch on an NVIDIA V100 GPU).

5 Results and Comparative Analysis

Table 2 reports the primary empirical findings on the 50-message Kazakh frozen test benchmark. Statistical confidence intervals were computed using Wilson score intervals.

Table 2: Performance Evaluation on the Kazakh Test Set (50 Messages)

Method	Scams Caught (out of 25)	False Alarms (out of 25)	Accuracy (%)	95% CI
Saq Heuristic Rules	2 / 25	0 / 25	54.0%	[40.4%, 67.0%
TF-IDF + LogReg	25 / 25	5 / 25	90.0%	[78.6%, 95.7%
Class Chat Model	24 / 25	0 / 25	98.0%	[89.5%, 99.7%
`scam-xlmr` (Fine-Tuned)	**25 / 25**	**2 / 25**	**96.0%**	**[86.5%, 98.9%**
XLM-R (Untrained Control)	25 / 25	25 / 25	50.0%	[36.8%, 63.2%

5.1 Effect of Local Kazakh Data Integration

Prior to integrating local Kazakh training examples, the XLM-R model trained exclusively on public English data detected 20 out of 25 Kazakh scam messages (80.0% sensitivity). Incorporating the localized Kazakh dataset significantly improved detection, raising scam retrieval to 25 out of 25 (100.0% sensitivity). False alarms on safe Kazakh messages shifted slightly from 1 to 2 out of 25, resulting in an overall Kazakh accuracy of 96.0%.

5.2 Comparison with Large Chat Models

The fine-tuned `scam-xlmr` model achieved 96.0% accuracy (250/250 scams caught, 2 false alarms), demonstrating competitive performance against the large generative chat model (98.0% accuracy, 24/25 scams caught, 0 false alarms). While the generative model slightly reduced false positives, `scam-xlmr` operates as a lightweight, specialized classifier that runs locally without dependency on external third-party APIs.

6 Discussion

In response to **RQ1**, cross-lingual transfer from public English data enables initial threat detection capabilities in Kazakh, supporting the premise that underlying social intent and deceptive structures transcend language boundaries [3]. Addressing **RQ2**, adding localized Kazakh instances is essential for capturing language-specific phrasing and regional threat vectors. Finally, regarding **RQ3**, specialized fine-tuning yields a model that performs comparably to large commercial chat architectures while maintaining deployment efficiency.

7 Limitations

Our evaluation is subject to the following limitations:

1. The test sample size ($N = 50$) means that each individual message accounts for 2 percentage points of total accuracy.
2. Experiments were evaluated using a single fixed decision threshold (0.5).
3. Safe test messages consist primarily of conversational dialogue rather than official transactional notifications.
4. Kazakh scam training and evaluation examples share sources from public threat bulletins.
5. Synthetic Kazakh training samples generated via AI prompts may introduce subtle distribution artifacts.
6. Public threat advisories could theoretically exist within the pre-training data of commercial chat models.

8 Availability and Web Demonstration

All source code, training configurations, and experimental scripts are open-sourced at: `https://github.com/dilya265/Saq`. We also developed a web demonstration interface for the **Saq** project, utilizing a color-coded traffic light system where Green indicates that *"no signs of fraud were detected"* (a recommendation rather than an absolute guarantee). Note that the web frontend currently operates on the heuristic rule suite.

9 Conclusion and Future Work

This work demonstrates that fine-tuning multilingual transformers with a small, localized Kazakh dataset provides highly effective scam detection. Future work will focus on expanding the Kazakh evaluation corpus (Test 2), developing model distillation techniques for mobile edge deployment, and integrating the fine-tuned `scam-xlmr` model directly into live messaging endpoints.

References

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